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Workshop-4

1. Read in the text file called numbers.txt and print out all the data to the terminal. You should also print out how many numbers are within the file.

Ans :

Source Code:

```
#include <stdio.h>

int main(){
 FILE *fp;
 int c,count=0;

 fp=fopen("Numbers.txt","r");
 while((c=fgetc(fp))!=EOF){
  printf("%d ",c);
  count++;
 }
 printf("\n Count = %d.",count);

 printf("\n");
 return 0;
}
```

Output:

```
54 54 51 10 50 54 10 52 56 10 54 55 10 54 56 10 54 56 10 49 49 10 52 50 10 52 56 10 54 49 10 54 48 10
54 56 10 53 48 10 51 57 10 53 51 10 57 54 10 56 54 10 52 49 10 54 52 10 55 51 10 52 53 10 53 49 10 49
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57 50 10 55 57 10 52 55 10 50 54 10 49 49 10 49 10 50 53 10 53 53 10 54 50 10 53 49 10 54 49 10 54 53
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56 50 10 56 53 10 55 52 10 57 54 10 49 56 10 54 56 10 56 52 10 55 49 10 50 54 10 52 56 10 55 55 10 52
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10 53 48 10 53 56 10 52 50 10 51 51 10 54 54 10 51 50 10 55 55 10 54 53 10 51 51 10 54 57 10 54 57 10
57 56 10 54 56 10 55 54 10 50 49 10 54 10 54 48 10 51 10 50 10 55 57 10 50 56 10 49 55 10 53 53 10 55
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55 54 10 56 55 10 57 48 10 57 56 10 51 51 10 57 54 10 53 57 10 57 53 10 55 49 10 52 56 10 56 56 10 49
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10 57 57 10 50 49 10 51 55 10 53 53 10 50 50 10 56 55 10 56 49 10 49 48 48 10 54 52 10 52 53 10 57 10
53 55 10 52 54 10 56 48 10 57 10 54 55 10 54 52 10 56 53 10 49 51 10 51 53 10 57 48 10 53 10 54 54 10
50 10 53 10 51 57 10 53 51 10 53 50 10 50 50 10 51 50 10 51 55 10 51 10 55 55 10 55 50 10 52 53 10 51
55 10 54 52 10 54 48 10 55 56 10 52 55 10 52 50 10 51 56 10 55 48 10 50 52 10 53 55 10 56 50 10 54 56
10 57 57 10 54 55 10 57 50 10 49 57 10 53 50 10 57 48 10 54 55 10 51 53 10 54 57 10 51 55 10 49 55 10
52 49 10 50 55 10 51 56 10 55 55 10 52 10 52 57 10 50 57 10 52 51 10 56 48 10 52 48 10 57 52 10 51 50
10 51 56 10 50 55 10 54 53 10 55 53 10 49 10 49 56 10 54 56 10 57 48 10 54 53 10 53 56 10 52 51 10 50
50 10 49 56 10 55 55 10 49 55 10 50 55 10 57 53 10 57 48 10 53 57 10 57 57 10 52 52 10 50 53 10 51 57
10 53 55 10 52 10 52 50 10 54 48 10 50 54 10 51 52 10 55 49 10 55 48 10 55 51 10 57 56 10 55 54 10 54
50 10 53 52 10 52 51 10 57 54 10 56 56 10 54 52 10 51 53 10 50 51 10 49 48 48 10 53 53 10 50 54 10 50
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54 10 54 52 10 54 57 10 51 49 10 57 53 10 54 48 10 51 56 10 57 54 10 54 55 10 53 50 10 50 55 10 57 50
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55 54 10 49 56 10 53 51 10 56 51 10 54 56 10 53 56 10 56 10 53 50 10 55 50 10 57 10 55 10 52 50 10 55
50 10 51 56 10 54 52 10 52 48 10 53 54 10 55 52 10 53 53 10 52 57 10 52 50 10 56 56 10 54 50 10 49 56
10 55 49 10 57 52 10 57 52 10 52 57 10 49 49 10 50 57 10 54 51 10 49 49 10 54 49 10 53 48 10 55 51 10
57 53 10 53 10 49 49 10 57 49 10 56 10 53 51 10 52 53 10 56 51 10 53 49 10 55 56 10 51 50 10 49 52 10
55 57 10 50 10 54 48 10 56 48 10 52 49 10 55 52 10 50 56 10 53 52 10 56 48 10 50 10 53 55 10 50 54 10
57 51 10 49 48 10 55 56 10 56 57 10 52 57 10 56 56 10 55 10 57 10 53 55 10 52 54 10 53 52 10 57 57 10
50 54 10 49 10 57 56 10 56 48 10 49 56 10 53 50 10 49 10 54 54 10 55 50 10 53 55 10 56 48 10 49 48 48
10 56 52 10 56 53 10 57 57 10 49 57 10 49 48 48 10 52 53 10 55 54 10 51 51 10 50 57 10 57 51 10 53 53
10 50 52 10 55 52 10 54 53 10 49 48 48 10 57 54 10 57 55 10 49 51 10 52 52 10 56 57 10 49 55 10 52 50
10 56 51 10 53 56 10 52 51 10 52 54 10 49 48 48 10 54 50 10 50 55 10 55 10 49 51 10 53 57 10 51 54 10
54 57 10 50 10 56 48 10 55 49 10 57 53 10 54 55 10 50 57 10 51 52 10 52 51 10 49 49 10 49 54 10 50 51
10 57 51 10 51 53 10 57 53 10 53 56 10 56 54 10 55 53 10 56 48 10 51 52 10 54 57 10 56 55 10 54 54 10
53 55 10 49 53 10 57 10 53 49 10 53 57 10 55 55 10 52 55 10 53 10 54 50 10 56 57 10 51 53 10 54 49 10
55 57 10 55 49 10 53 50 10 52 52 10 51 10 49 49 10 54 54 10 53 55 10 53 48 10 51 51 10 49 51 10 52 55
10 54 54 10 50 10 52 54 10 57 56 10 50 52 10 55 53 10 53 49 10 55 49 10 51 10 52 57 10 49 48 48 10 49
56 10 51 53 10 49 53 10 50 51 10 53 10 49 48 48 10 50 55 10 51 48 10 53 54 10 57 56 10 49 51 10 55 10
55 48 10 56 56 10 56 49 10 50 51 10 49 57 10 49 10 52 50 10 56 53 10 51 49 10 53 49 10 54 53 10 55 57
10 53 50 10 55 52 10 52 54 10 49 57 10 51 49 10 57 54 10 51 52 10 57 51 10
 Count = 1895.
```

2. Read in the text file called numbers.txt and detect all even numbers.

Ans :

Source Code:

```c
#include<stdio.h>

int main(){
 FILE *fp;
 int num,count=0;

 fp=fopen("Numbers.txt","r");
 while(fscanf(fp,"%d",&num)!=EOF){
  if(num%2==0){
   printf("%d ",num);
   count++;
  }
 }
 printf("\nThere are %d even numbers in the file.",count);
 fclose(fp);

 printf("\n");
 return 0;
}
```

Output:

```
26 48 68 68 42 48 60 68 50 96 86 64 16 2 94 26 82 36 58 92 26 62 64 6
68 62 12 98 62 44 86 34 24 28 32 64 72 34 4 82 8 54 100 54 92 42 90 82
 74 96 18 68 84 26 48 6 80 82 8 36 36 82 76 38 50 58 42 66 32 98 68 76
 6 60 2 28 78 58 24 74 20 86 78 34 92 18 22 6 54 28 18 86 78 48 70 44
22 76 74 100 90 80 50 44 50 70 92 56 28 20 66 84 76 90 98 96 48 88 70
44 28 56 88 100 42 22 100 64 46 80 64 90 66 2 52 22 32 72 64 60 78 42
38 70 24 82 68 92 52 90 38 4 80 40 94 32 38 18 68 90 58 22 18 90 44 4
42 60 26 34 70 98 76 62 54 96 88 64 100 26 20 56 40 54 10 62 40 52 54
56 28 70 34 96 64 60 38 96 52 92 24 68 100 70 56 6 70 36 70 90 38 48 3
0 16 22 12 76 70 26 44 26 62 78 66 76 18 68 58 8 52 72 42 72 38 64 40
56 74 42 88 62 18 94 94 50 8 78 32 14 2 60 80 74 28 54 80 2 26 10 78 8
8 46 54 26 98 80 18 52 66 72 80 100 84 100 76 24 74 100 96 44 42 58 46
 100 62 36 2 80 34 16 58 86 80 34 66 62 52 44 66 50 66 2 46 98 24 100
18 100 30 56 98 70 88 42 52 74 46 96 34
There are 319 even numbers in the file.
```

3. Reproduce number 2 but instead, print all the numbers to a new text file.

Ans :

Source Code:

```c
#include<stdio.h>

int main(){
 FILE *fp1,*fp2;
 int num,count=0;

 fp1=fopen("Numbers.txt","r");
 fp2=fopen("even-numbers.txt","w+");


 while(fscanf(fp1,"%d",&num)!=EOF){
  if(num%2==0){
   fprintf(fp2,"%d\n",num);
  }
 }
 rewind(fp2);
 while(fscanf(fp2,"%d",&num)!=EOF){
  if(num%2==0){
   printf("%d ",num);
   count++;
  }
 }

 printf("\nThere are %d even numbers in the file.",count);
 fclose(fp1);
 fclose(fp2);

 printf("\n");
 return 0;
}
```

Output:

```
26 48 68 68 42 48 60 68 50 96 86 64 16 2 94 26 82 36 58 92 26 62 64 6
68 62 12 98 62 44 86 34 24 28 32 64 72 34 4 82 8 54 100 54 92 42 90 82
 74 96 18 68 84 26 48 6 80 82 8 36 36 82 76 38 50 58 42 66 32 98 68 76
 6 60 2 28 78 58 24 74 20 86 78 34 92 18 22 6 54 28 18 86 78 48 70 44
22 76 74 100 90 80 50 44 50 70 92 56 28 20 66 84 76 90 98 96 48 88 70
44 28 56 88 100 42 22 100 64 46 80 64 90 66 2 52 22 32 72 64 60 78 42
38 70 24 82 68 92 52 90 38 4 80 40 94 32 38 18 68 90 58 22 18 90 44 4
42 60 26 34 70 98 76 62 54 96 88 64 100 26 20 56 40 54 10 62 40 52 54
56 28 70 34 96 64 60 38 96 52 92 24 68 100 70 56 6 70 36 70 90 38 48 3
0 16 22 12 76 70 26 44 26 62 78 66 76 18 68 58 8 52 72 42 72 38 64 40
56 74 42 88 62 18 94 94 50 8 78 32 14 2 60 80 74 28 54 80 2 26 10 78 8
8 46 54 26 98 80 18 52 66 72 80 100 84 100 76 24 74 100 96 44 42 58 46
 100 62 36 2 80 34 16 58 86 80 34 66 62 52 44 66 50 66 2 46 98 24 100
18 100 30 56 98 70 88 42 52 74 46 96 34
There are 319 even numbers in the file.
```

4. Using the command line arguments (argv), copy the contents of one file to another file. For example. If you run your program with "./a.out file1.txt file2.txt", your program should transfer the content from file1 to file2.

Ans :

Source Code:

```
#include<stdio.h>

int main(int argc,char *argv[]){
 FILE *fp1,*fp2;
 int c;

 if(argc!=3){
  printf("Usage: %s source_file destination_file\n",argv[0]);
  return 1;
 }

 fp1=fopen(argv[1],"r");
 if(fp1==NULL){
  printf("Error opening source file.\n");
  return 1;
 }

 fp2=fopen(argv[2],"w");
 if(fp2==NULL){
  printf("Error opening destination file.\n");
  fclose(fp1);
  return 1;
 }
 while((c=fgetc(fp1))!=EOF){
  fputc(c,fp2);
 }
 printf("File copied successfully");

 fclose(fp1);
 fclose(fp2);

 printf("\n");
 return 0;
}
```

Output:

```
praju-maharjan@LOCKON:~/Algorithms_and_Concurrency/Workshop-4$ ./Q4 Nu
mbers.txt Numbers2.txt
File copied successfully
```

5. Using the command line arguments (argv), merge the contents of two files and print them out to another. For example, when you run your program with ./a.out file1.txt file2.txt file3.txt, the contents from file1 and file2 will be combined and printed to file3. Both files you will need to create yourself, it

doesn't matter what is inside, just as long as you're able to see the merge at the end.

Ans :

Source Code:

```
#include<stdio.h>

int main(int argc,char *argv[]){
 FILE *fp1,*fp2,*fp3;
 int c;

 if(argc!=4){
  printf("Usage: %s sfile1 file2 combined_file\n",argv[0]);
  return 1;
 }

 fp1=fopen(argv[1],"r");
 if(fp1==NULL){
  printf("Error opening source file 1.\n");
  return 1;
 }

 fp2=fopen(argv[2],"r");
 if(fp2==NULL){
  printf("Error opening source file 2.\n");
  fclose(fp1);
  return 1;
 }

 fp3=fopen(argv[3],"w");
 if(fp3==NULL){
  printf("Error opening destination file.\n");
  fclose(fp1);
  fclose(fp2);
  return 1;
 }

 while((c=fgetc(fp1))!=EOF){
  fputc(c,fp3);
 }
 while((c=fgetc(fp2))!=EOF){
  fputc(c,fp3);
 }

 printf("File copied successfully");

 fclose(fp1);
 fclose(fp2);
 fclose(fp3);

 printf("\n");
 return 0;
}
```

Output:

```
praju-maharjan@LOCKON:~/Algorithms_and_Concurrency/Workshop-4$ ./Q5 Numbers.txt Numbers2.txt Numbers3.txt
File copied successfully
```

6. Using scanf, ask the user to type in their name and age, print this data to a text file. You should aim to only use one scanf command as you can store memory to an integer variable and char array using %s and %d.

Ans :

Source Code:

```
#include<stdio.h>

int main(){
 FILE *fp;
 char name[50];
 int age;

 printf("Enter name and age :");
 scanf("%s %d",name,&age);

 fp=fopen("name-age.txt","w");
 if(fp==NULL){
  printf("Error opening file.\n");
  return 1;
 }
 fprintf(fp,"Name : %s\nAge : %d\n",name,age);
 fclose(fp);

 printf("Name and age written to file successfully");

 printf("\n");
 return 0;
}
```

Output:

```
Enter name and age :Praju 20
Name and age written to file successfully
```

7. Read in Words.txt which contains a single paragraph, perform the following:
   a. Print out how many times the letter “c” is found in the text file.
   b. Print out how many words are in the text file.
   c. Print out how many punctuation symbols are used (, . - / etc)

Ans :

Source Code:

```c
#include <stdio.h>

int main(){
    FILE *fp;
    char ch;
    int c_count=0,word_count=0,p_count=0;
    int inWord=0;

    fp=fopen("Words.txt", "r");
    if(fp==NULL){
        printf("Error opening file.\n");
        return 1;
    }

    while((ch=fgetc(fp))!=EOF){
        if(ch=='c'||ch=='C'){
            c_count++;
        }

        if((ch>='A'&&ch<='Z')||(ch>='a'&&ch<='z')||(ch>='0' && ch<= '9')){
            if(inWord==0){
                word_count++;
                inWord=1;
            }
        }else{
            inWord=0;
        }
        if((ch >=33&&ch<=47)||
           (ch >=58&&ch<=64)||
           (ch >=91&&ch<=96)||
           (ch >=123&&ch<=126)){
            p_count++;
        }
    }

    fclose(fp);

    printf("Number of 'c' letters: %d\n",c_count);
    printf("Number of words: %d\n", word_count);
    printf("Number of punctuation symbols: %d\n",p_count);

    return 0;
}
```

Output:

```
Number of 'c' letters: 20
Number of words: 108
Number of punctuation symbols: 18
```

8. Read in Words.txt and encrypt all the characters by adding 3 to the ascii value. For example, the letter a will translate to d. The letter h will translate to k. You should then print this value to a text file.

Ans :

Source Code:

```c
#include <stdio.h>

int main(){
    FILE *fp1, *fp2;
    char ch;

    fp1 = fopen("Words.txt", "r");
    if(fp1 == NULL){
        printf("Error opening input file.\n");
        return 1;
    }

    fp2 = fopen("Encrypted.txt", "w");
    if(fp2 == NULL){
        printf("Error opening output file.\n");
        fclose(fp1);
        return 1;
    }

    while((ch = fgetc(fp1)) != EOF){
        ch = ch + 3;
        fputc(ch, fp2);
    }

    printf("File encrypted successfully.\n");

    fclose(fp1);
    fclose(fp2);

    return 0;
}
```

Output:

```
File encrypted successfully.
```